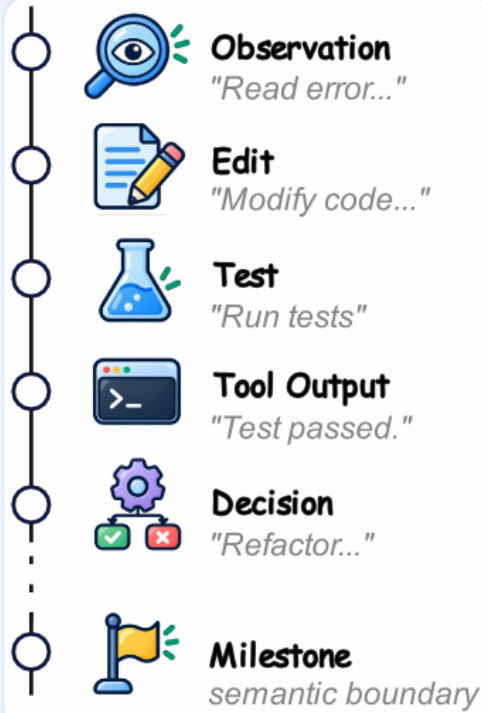


Structure Execution into Memory Traces

Execution History



WAL

Write-Ahead Log

- append-only
- ordered events
- durable

persist

Memory Trace



Memory Episode



Trace Store

Immutable
Never Overwritten

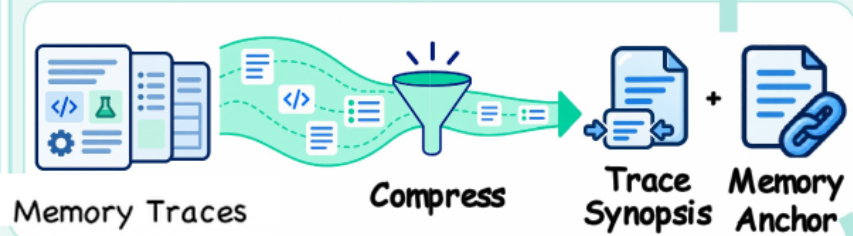


Addressable Working Context

Working Context



Memory Offloading



Memory Traces

Compress

Trace Synopsis

Memory Anchor

Persist full content externally

Memory Index

Anchor ID	Type	Section	Version	Metadata
m_001	trace	p_1-10	v3	parser.py
m_002	trace	p_20-28	v1	tests/
m_003	code	p_5-12	v2	utils.py
...	...	...	...	...



Resolve to trace

Memory trace Graph (MTG)

Execution Relations



updated with execution events

Repository State Graph (RSG)

Repository Structure



updated with repository state changes



Shared Repository Entity
File
Symbol
Test
...

reaching limit

replace

still full

Epoch limit reached
· Logical Fallback Exhausted
· Context Cannot Fit
· Thread / Provider fails

Create Checkpoint
· Current Task State
· Relevant Memory Traces
· Relevant Code Files
· Execution Frontier

New Epoch
· Task State
· Memory Traces
· Code Files
· Execution Frontier

Continue Execution
Seamless long-horizon progression!

Minimal Recall with Validation

Trigger Recall



Guided by
MTG and RSG

Memory Index Translation

① Memory Trace

Trace: T17

2025-01-12 10:24
Fix failing test
sklearn/svm/
Read files
Edit code
Run tests
Error logs
Discussion
...

② Section

Section: Implementation

Modify decision_function in base.py
Edit hunks (diff)
Run related tests
...

Metadata
type: code_modification
files: sklearn/svm/base.py

③ Evidence Range

Range: R3(base.py)

```
768 def decision_function(self, X):
769     if self.probability:
770         return self._predict_proba_lr(X)
771     # fixed: handle empty input
772     if X.size == 0:
773         return np.array([])
774     return self._decision_function(X)
...
```

④ Minimal Evidence

Lines: 771-773

```
771 ## fixed: handle empty input
772 +if X.size == 0:
773 + return np.array([])
```

Validated against
Repository State and Task State

Validated
Reload

Inject to
Working Context